# Factual History Visual Learning Engine: Architecture Spec

### TECHNICAL IMPLEMENTATION SPECIFICATION FOR ANTIGRAVITY • LONGFORM & SHORTS REFACTOR

This document specifies the exact code architecture, data contracts, and implementation steps required to transform the existing `yt-automation` codebase into a high-retention, factual interactive documentary engine. It adds primary source verification, animated tactical maps, dynamic visual chapter progress markers, and dual-mode pipeline orchestration.

## 1. System Architecture Overview & Module Map

The architecture transitions the generation flow from speculative counterfactuals to a rigorous 4-layer documentary synthesis stack. The diagram below details the data flow between modules:

```
[User Topic / Historical Event]
      ↓
[Stage 1: Primary Source Ingestion & RAG Verifier] (pipeline/primary_sources.py)
      • Academic Consensus & Primary Eyewitness Ingestion (Diaries, Tablets, Treaties)
      • On-Screen Manuscript Translation Card Generation
      ↓
[Stage 2: Script Architect & Visual Scene Planner] (longform_engine/script_architect.py)
      • 7-Act Dramatic Structure (80-85 visual beats, 7.5s average cut)
      • Scene Taxonomy: 45% Cinematic Wide, 25% Tactical Map, 15% Manuscript, 15% Macro
      ↓
[Stage 3: Multi-Layer Asset Rendering Engine]
      • Layer A: Fooocus SDXL + Negative Prompt Lock (Photorealistic Keyframes)
      • Layer B: TacticalMapEngine (Animated marching vectors & elevation maps via
OpenCV)
      • Layer C: Edge-TTS Multi-Language Audio Synthesis (EN, ES, PT, DE, HI)
      ↓
[Stage 4: Compositor & Interactive Overlay Engine]
(longform_engine/interactive_overlays.py)
      • 4-Trajectory Smoothstep Ken-Burns Motion Engine
      • Top-Right Dynamic Act Progress Badges ("Act II: The Overland Fleet • 1453 AD")
      • Subtitles: Long-form 16:9 Lower-Third (MarginV=120) • Short-form 9:16 Safe-Zone
(MarginV=620)
      • 3-Layer Audio Ducking: Voiceover (0dB), Sound FX (-8dB), Cinematic Bed (-18dB)
      ↓
[Stage 5: Multi-Platform Funnel Distributor] (longform_engine/short_splitter.py)
      • 16:9 Master Documentary MP4 (1080p, 10–12 mins)
      • 5x 9:16 Vertical Shorts with Embedded "Related Video" Funnel Metadata
```

## 2. Technical Component Specifications

### Component A: Primary Source Ingestion Layer (`pipeline/primary_sources.py`)

Replaces speculative counterfactual generation with strict historiographical grounding. Extracts verbatim historical quotes, dates, and archival citations to burn into on-screen translation callouts.

| Method / Interface | Input Schema | Output Contract & Behavior |
|---|---|---|
| `extract_primary_sources(event_slug: str)` | `event_slug`: e.g. `"fall_of_constantinople_1453"` | Returns list of validated source objects containing: `author`, `date`, `original_work`, `quote`, `translation`, and `academic_citation`. |
| `render_source_card(source_dict, out_png)` | `source_dict`: Single source record<br>`out_png`: Target output path | Renders a 1920×1080 transparent PNG with an aged parchment text box, author title, and highlighted translated quote. |

### Component B: Tactical Map Engine (`longform_engine/tactical_map_renderer.py`)

Renders 16:9 animated tactical maps illustrating troop movements, naval routes, and fortress sieges using OpenCV, Pillow, and matplotlib. Eliminates static image boredom by providing visual geography.

| Function | Parameters | Visual Output |
|---|---|---|
| `render_tactical_siege_map(...)` | `base_map_path`: Topographic map<br>`vectors`: List of `[x1, y1, x2, y2, color, label]`<br>`duration_sec`: 6.0–10.0s | Generates an MP4 clip with animated dotted movement arrows, pulsing battle markers, and topographic zoom. |
| `render_overland_fleet_transit(...)` | `elevation_profile`: Galata ridge coords<br>`ship_count`: 70 | Renders a visual timeline animation tracking naval ships traversing overland contours from the Bosporus to the Golden Horn. |

### Component C: Interactive Visual Overlay Engine (`longform_engine/interactive_overlays.py`)

Burns dynamic progress indicators and primary source callouts directly onto video clips without third-party editing software.

- **Upper-Right Act Progress Badge:** Renders a subtle semi-transparent badge at `(X=1520, Y=60)` displaying `Act Number`, `Act Title`, and `Historical Date`. Fades in at act starts (0.5s fade) and fades out at act conclusions.
- **Document Zoom Callout:** Overlays high-resolution manuscript scans with animated amber bounding boxes highlighting key translated passages.
- **Sidechain Audio Compressor:** Employs FFmpeg's `sidechaincompress` filter with `threshold=0.08`, `ratio=6`, and `release=250ms` to dynamically dip background orchestral score under voiceover narration.

## 3. Antigravity Step-by-Step Implementation Checklist

Execute the following implementation plan in a single iteration to complete the upgrade:

1. **Create `pipeline/primary_sources.py`:** Implement `extract_primary_sources()` with embedded verified corpora for Constantinople (1453), Bronze Age Collapse (1177 BC), Cambyses (524 BC), and Operation Mincemeat (1943).
2. **Create `longform_engine/tactical_map_renderer.py`:** Add OpenCV and Pillow vector drawing routines to animate arrows and territory highlights across 16:9 map backgrounds.
3. **Create `longform_engine/interactive_overlays.py`:** Implement `burn_act_badge()` and `burn_source_card()` using FFmpeg filtergraphs (`overlay`, `drawtext`).
4. **Upgrade `longform_engine/assemble_longform.py`:** Wire `tactical_map_renderer` into shot types labeled `"Tactical Map"` and `"Siege Map"`, replacing standard static images with animated tactical clips.
5. **Add Dual-Mode Flag to `longform_engine/script_architect.py`:** Support `--mode factual` (academic consensus & primary sources) and `--mode counterfactual` (divergence chains).
6. **Run Unit Tests & Validation:** Add `tests/test_factual_learning_engine.py` validating map generation, source rendering, and overlay compilation.